Chorer: a tool for Program Understanding [1]

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This tool could help you debug your program!

The idea

We wanted to build a tool that could assists developers in analyzing existing codebases by applying **Choreography Automata** [2] theory to uncover classic bugs in distributed systems, particularly in the context of *actor-based programming*. To achieve this, our extraction algorithm follows a **bottom-up approach** with **over-approximation**, ensuring that every possible behavior (good or bad) is highlighted.

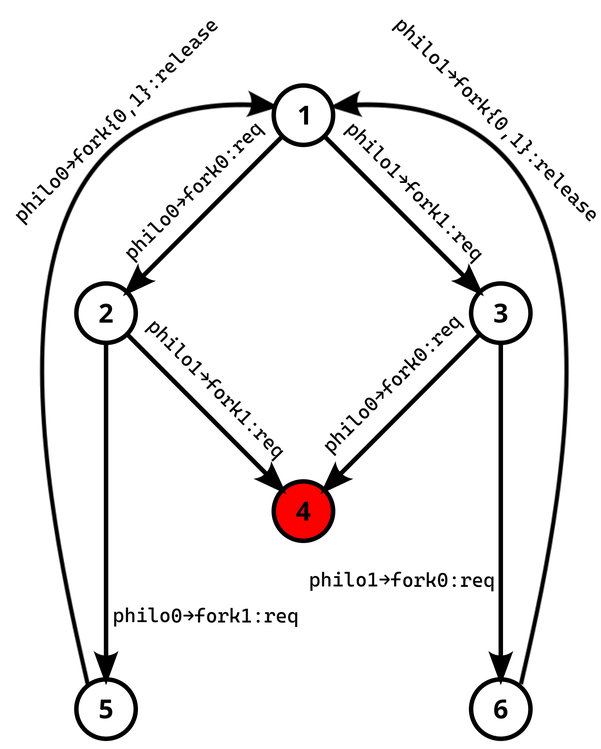
Extraction steps

1. Take the file input and perform some simple analysis
2. Create a local-view for each actor found
3. Create a global-view using the local-views
4. Minimize and analyse the output graph

Pseudocode

```
philosopher(Fork1, Fork2) →
  send req to Fork1,
  receive ack from Fork1,
  send req to Fork2,
  receive ack from Fork2,
  eat(),
  send release to Fork1,
  send release to Fork2,
  philosopher(Fork1, Fork2).

fork() →
  receive req from Phil,
  send ack to Phil,
  receive release from Phil,
  fork().
```



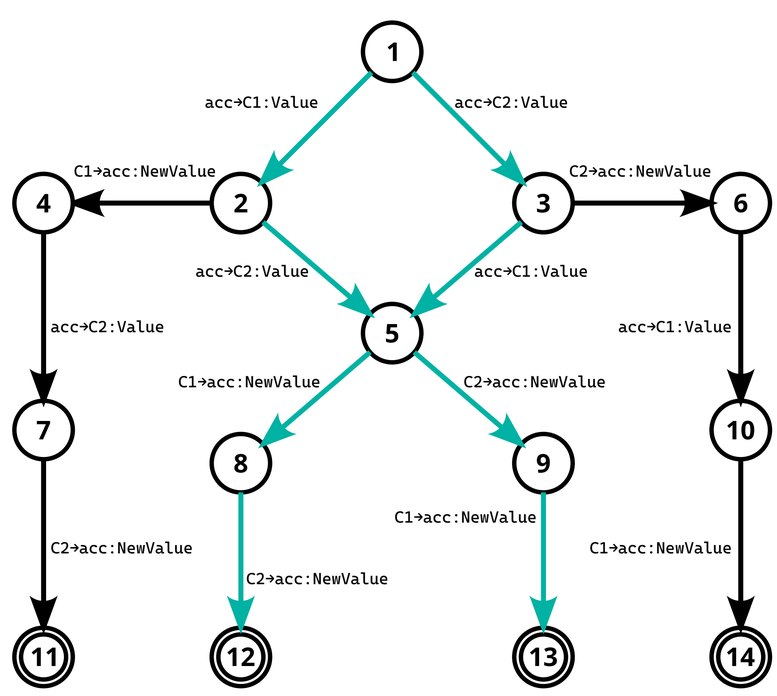
A motivating example

On the left, you can see an implementation of the **dining philosophers** problem written in *Erlang-like* pseudocode with its corresponding Choreography Automaton extracted by the tool. The system consists of four actors: two philosophers and two forks. The red state highlights a **deadlock** situation, which occurs when both philosophers attempt to take the forks simultaneously.

Pseudocode

```
account(Value) →
  receive
    read from Client →
      send Value to Client,
      account(Value),
    NewValue from Client →
      account(NewValue).

client() →
     receive read from Acc,
     receive Value from Acc,
     % operation on Value
     send NewValue to Acc.
```



Another example

We now demonstrate how the tool can highlight a **race condition**. Suppose a *banking application* attempts to read a client's credit information. If two types of client actors exist (let's call them *C1* and *C2*) the system may encounter a race condition. This potential behavior is illustrated by the light blue path in the automaton.

Conclusion

Building an effective tool that truly achieves its goals involves several challenges. The problem is inherently **undecidable**, as it would require solving termination (an already known undecidable problem). Moreover, the extraction process can produce extremely **large automata**. Significant effort has been dedicated to ensuring a correct and practical synthesis of the automaton.

Referance

[1] Genovese, et al. "Choreographies for Program Understanding", International Conference on Formal Techniques for Distributed Objects, Components, and Systems. 2025.

[2] Barbanera, et al. "Choreography automata" International Conference on Coordination Languages and Models. Cham: Springer International Publishing, 2020.

Repository

github.com/gabrielegenovese/chorer

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